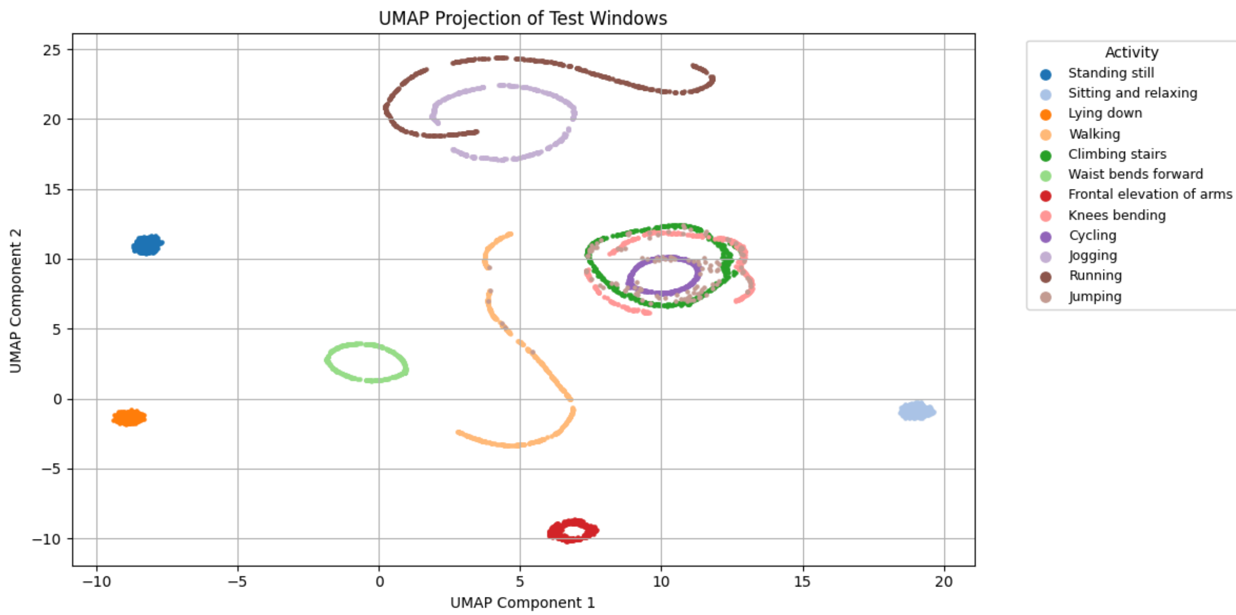
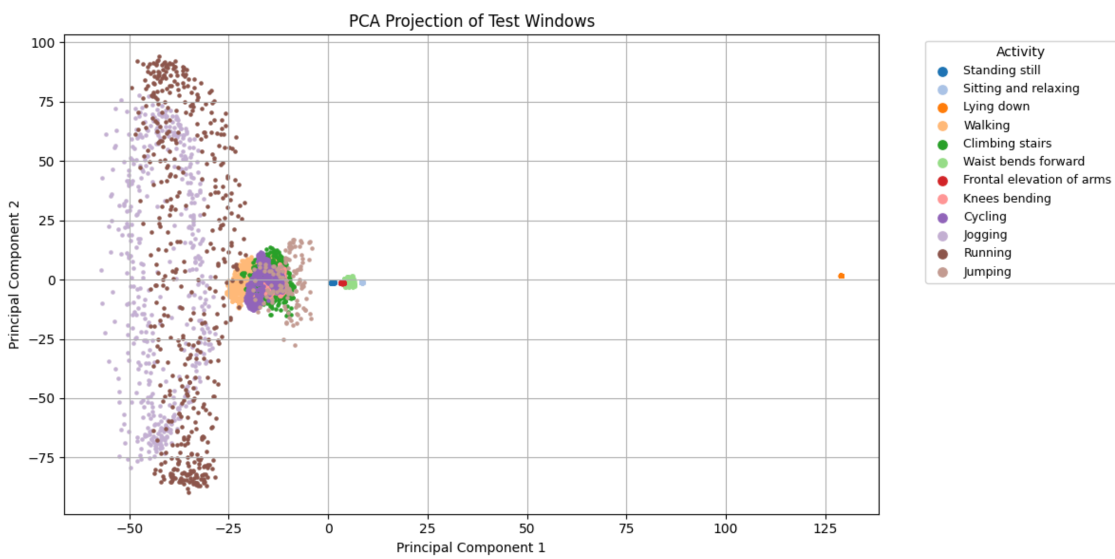
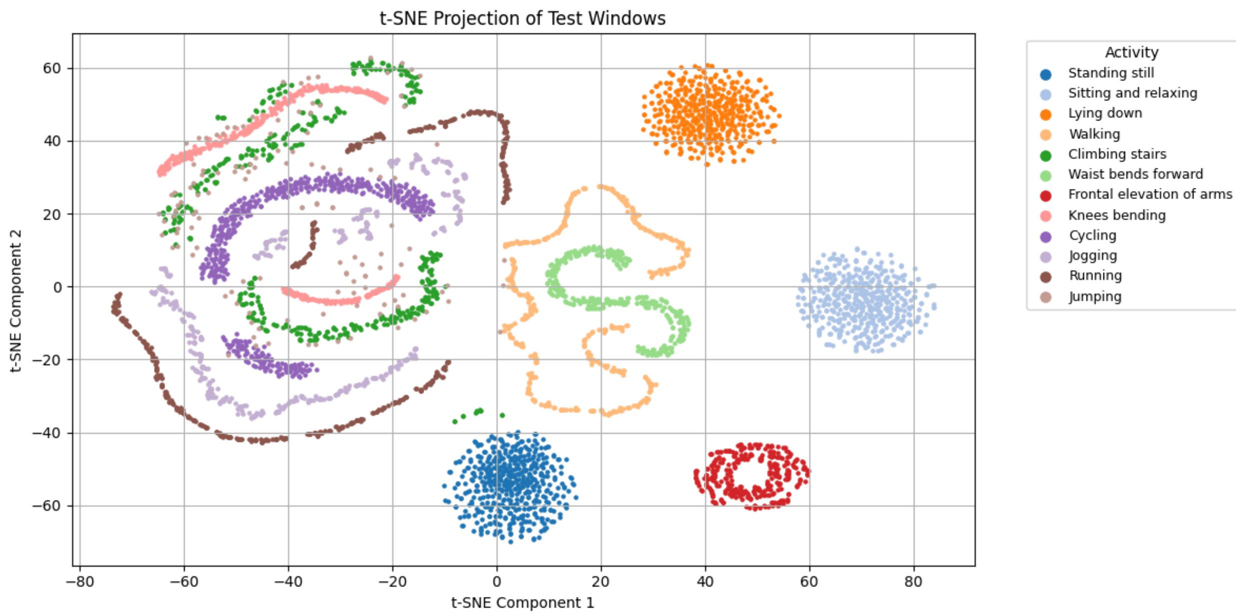

UMAP Projection of Test Windows
UMAP Component 1
UMAP Component 2
−10
−5
0
5
10
15
20
25
Activity
Standing still
Sitting and relaxing
Lying down
Walking
Climbing stairs
Waist bends forward
Frontal elevation of arms
Knees bending
Cycling
Jogging
Running
Jumping

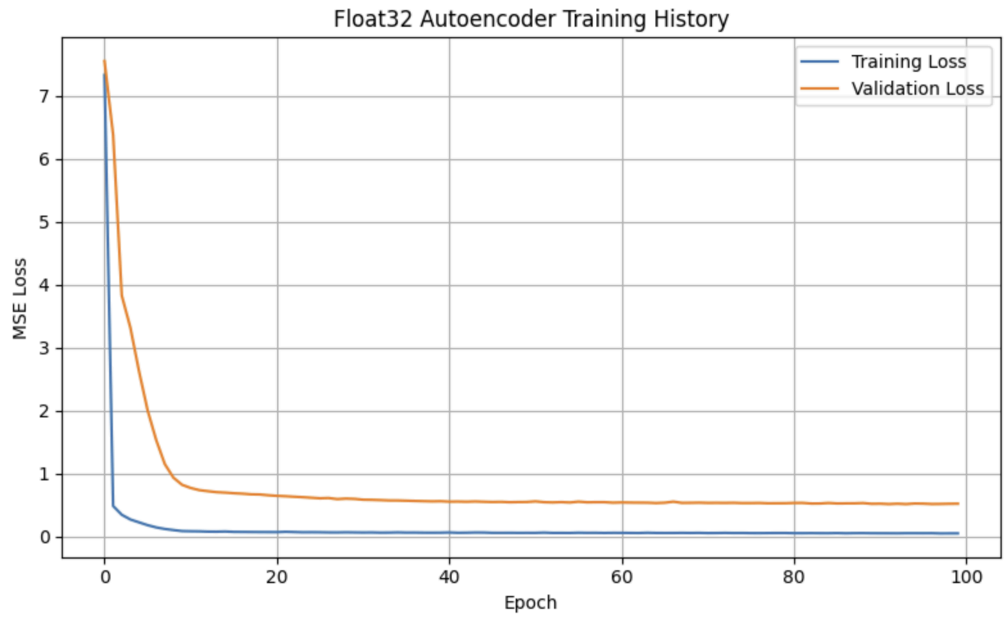

PCA Projection of Test Windows
Principal Component 1
Principal Component 2
−75
−50
−25
0
25
50
75
100
125
Activity
Standing still
Sitting and relaxing
Lying down
Walking
Climbing stairs
Waist bends forward
Frontal elevation of arms
Knees bending
Cycling
Jogging
Running
Jumping

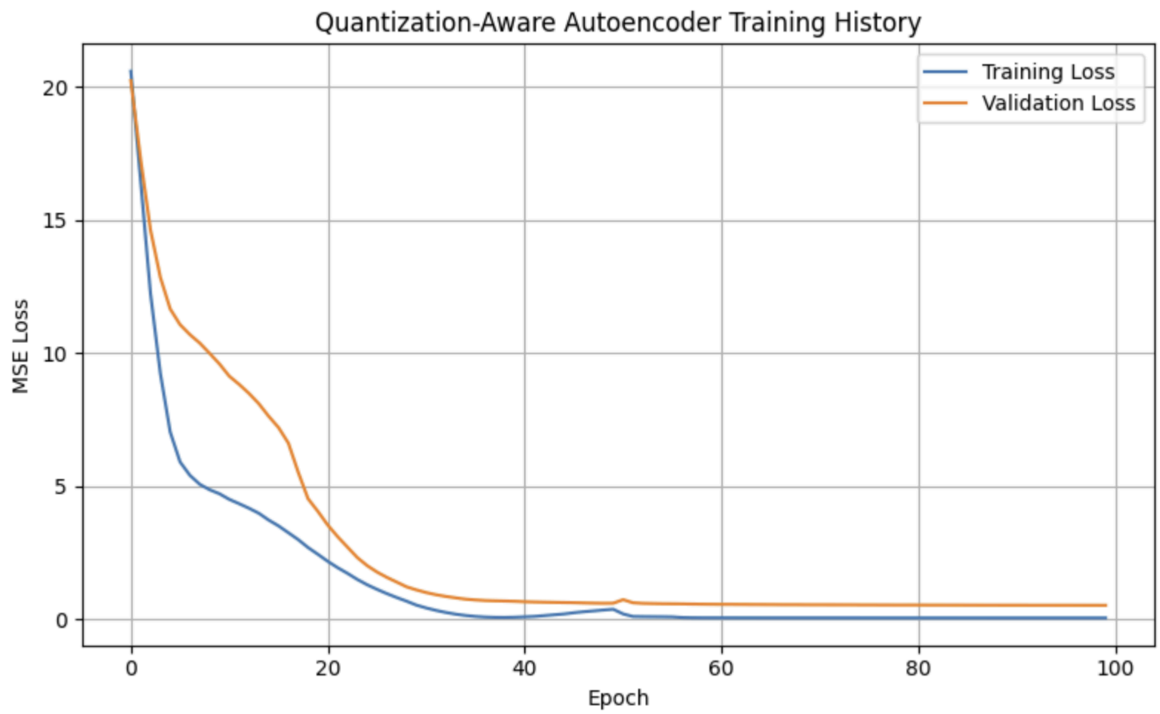

t-SNE Projection of Test Windows
t-SNE Component 1
t-SNE Component 2
Activity
Standing still
Sitting and relaxing
Lying down
Walking
Climbing stairs
Waist bends forward
Frontal elevation of arms
Knees bending
Cycling
Jogging
Running
Jumping
−80
−60
−40
−20
0
20
40
60
80
−60
−40
−20
0
20
40
60

Model: "anomaly_autoencoder"

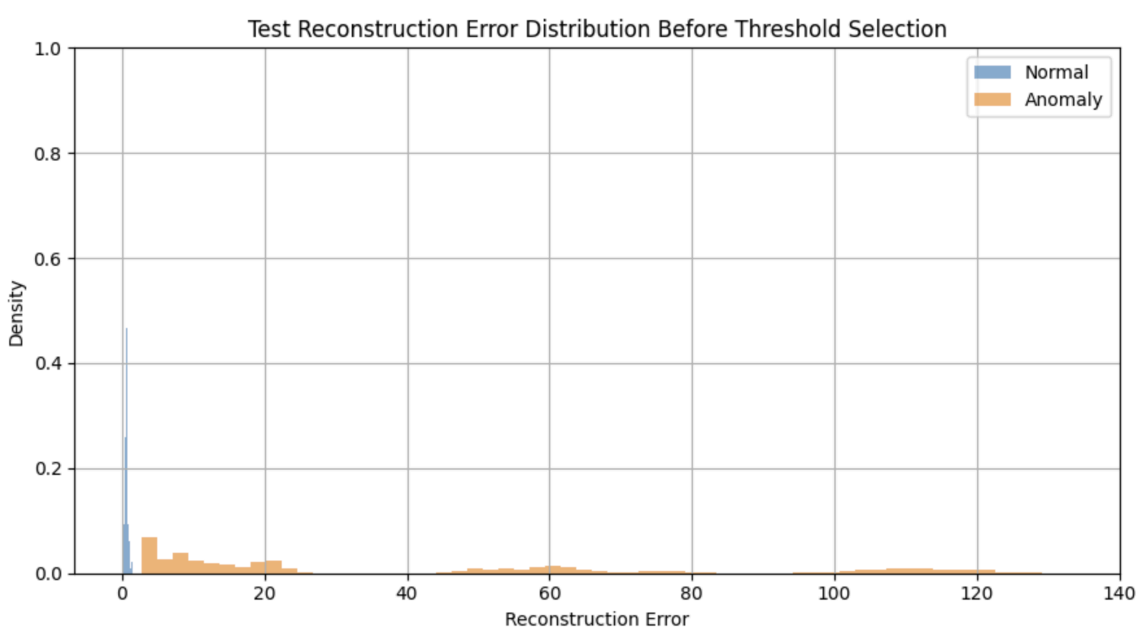
Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 300)	0
dense (Dense)	(None, 32)	9,632
dense_1 (Dense)	(None, 16)	528
dense_2 (Dense)	(None, 32)	544
dense_3 (Dense)	(None, 300)	9,900

...






```
Train normal      mean=0.099251, stderr=0.001688, min=0.008352, max=1.091108
Train anomaly     mean=41.639881, stderr=0.364598, min=2.473858, max=130.010788
Test normal       mean=0.138976, stderr=0.003998, min=0.009312, max=1.401435
Test anomaly      mean=42.055332, stderr=0.611826, min=2.806751, max=133.418747
```



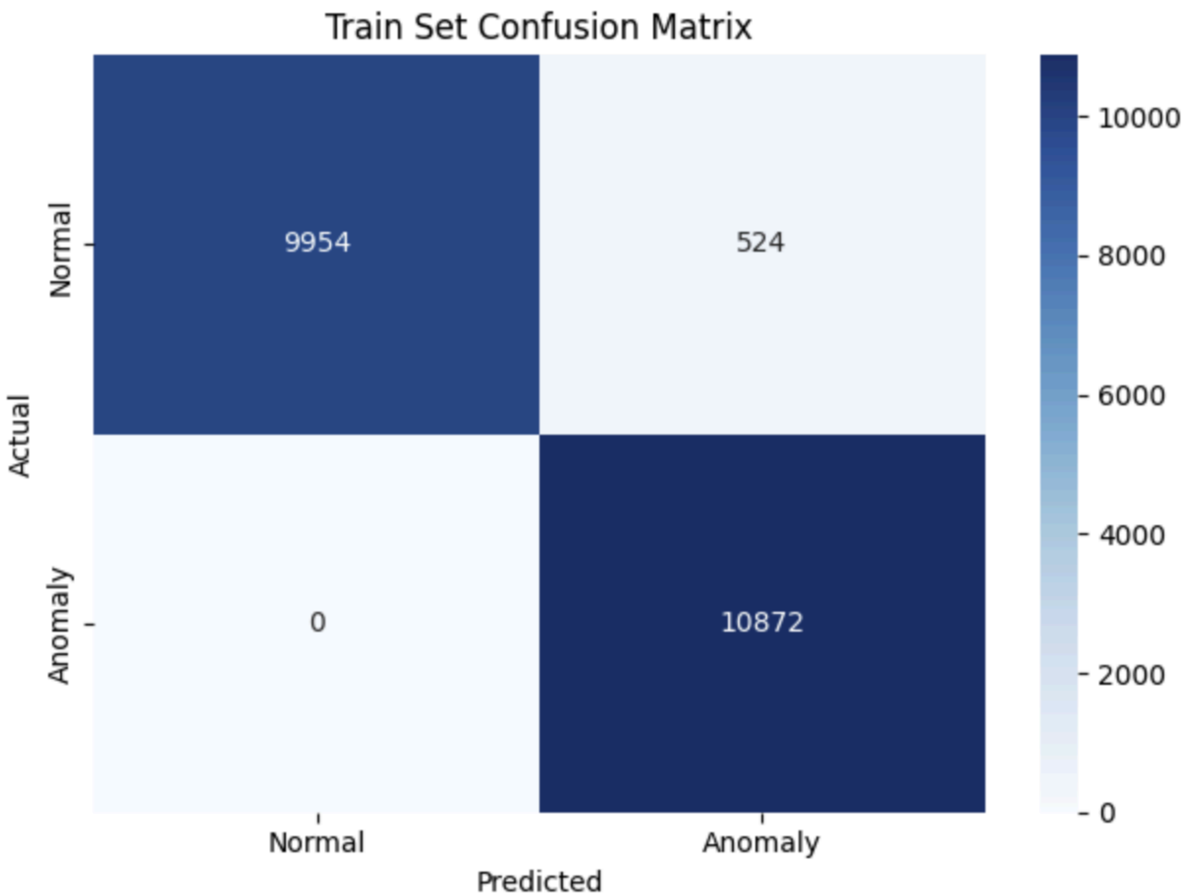

```
Selected threshold (95th pct of train normal errors): 0.508632
```

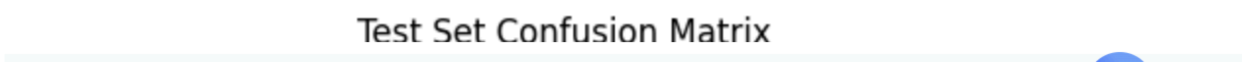
Train set evaluation

	precision	recall	f1-score	support
Normal (0)	1.00	0.95	0.97	10478
Anomaly (1)	0.95	1.00	0.98	10872
accuracy			0.98	21350
macro avg	0.98	0.97	0.98	21350
weighted avg	0.98	0.98	0.98	21350

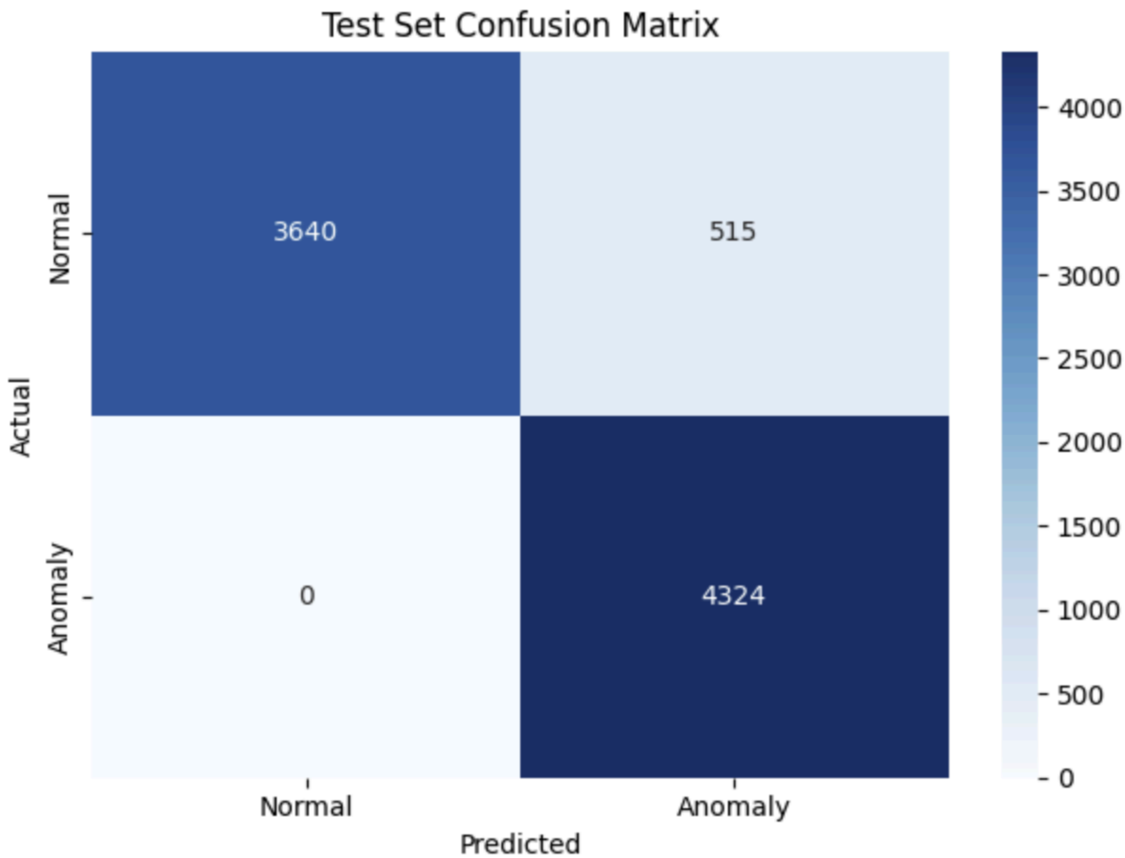
Test set evaluation

	precision	recall	f1-score	support
Normal (0)	1.00	0.88	0.93	4155
Anomaly (1)	0.89	1.00	0.94	4324
accuracy			0.94	8479
macro avg	0.95	0.94	0.94	8479
weighted avg	0.95	0.94	0.94	8479

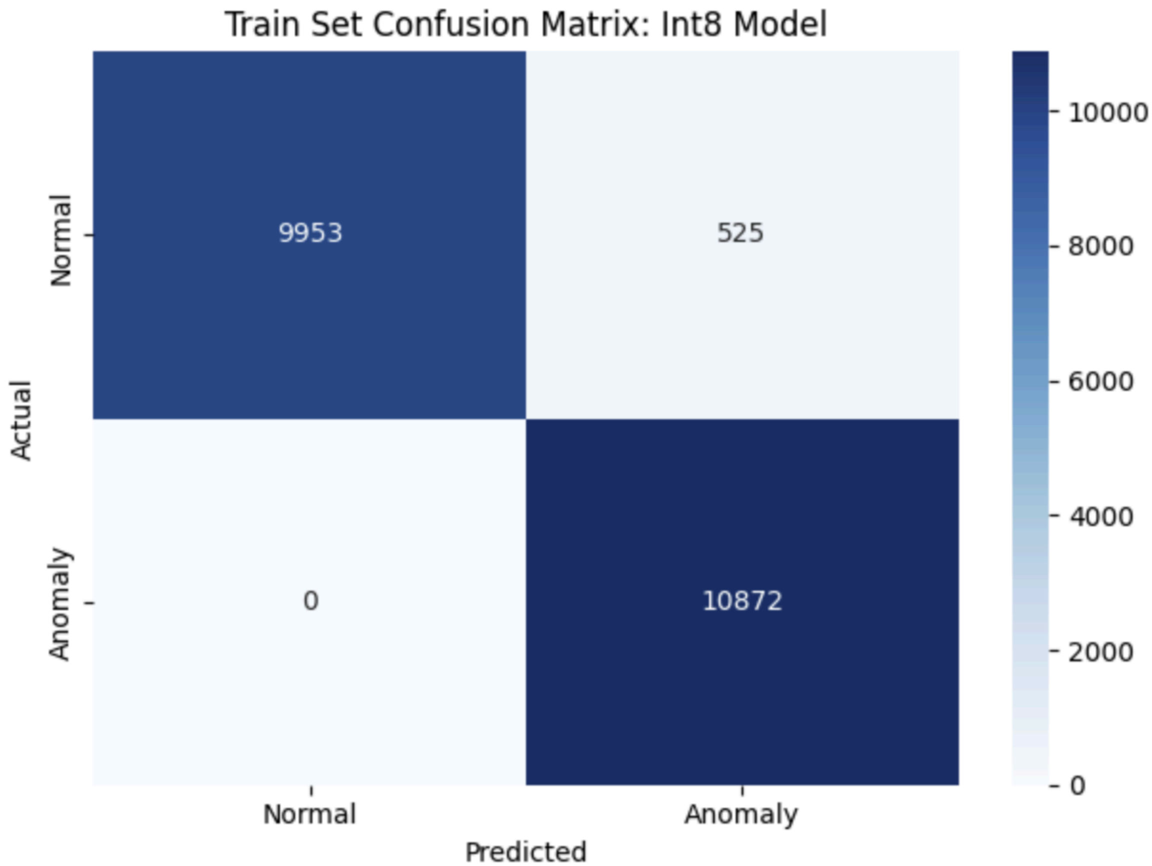

Train Set Confusion Matrix
Normal
Anomaly
Actual
9954
524
0
10872
Normal
Anomaly
Predicted
10000
8000
6000
4000
2000
0

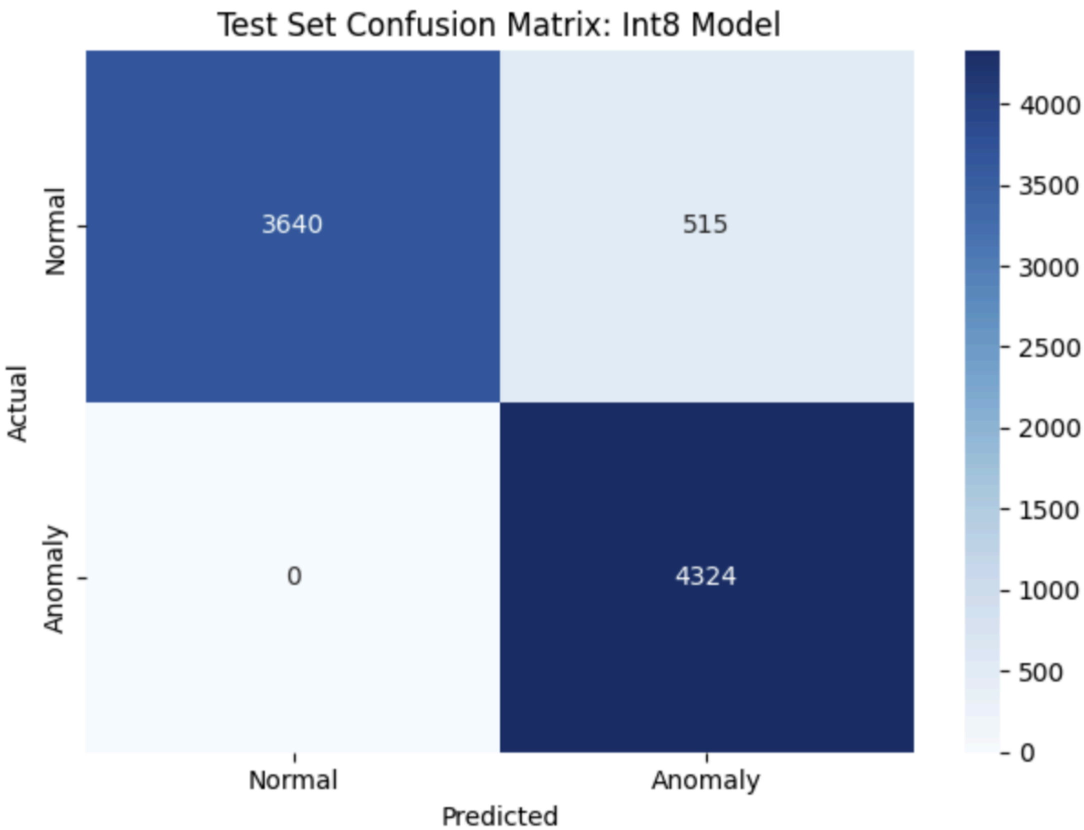

Test Set Confusion Matrix

•


Test Set Confusion Matrix
Actual
Normal
Anomaly
3640
515
0
4324
Predicted
Normal
Anomaly
4000
3500
3000
2500
2000
1500
1000
500
0

```
Train set evaluation: int8 model
              precision    recall  f1-score   support

  Normal (0)       1.00      0.95      0.97     10478
 Anomaly (1)       0.95      1.00      0.98     10872

    accuracy                           0.98     21350
   macro avg       0.98      0.97      0.98     21350
weighted avg       0.98      0.98      0.98     21350

Test set evaluation: int8 model
              precision    recall  f1-score   support

  Normal (0)       1.00      0.88      0.93      4155
 Anomaly (1)       0.89      1.00      0.94      4324

    accuracy                           0.94      8479
   macro avg       0.95      0.94      0.94      8479
weighted avg       0.95      0.94      0.94      8479
```


Train Set Confusion Matrix: Int8 Model
Actual
Normal
Anomaly
9953
525
0
10872
Normal
Anomaly
Predicted
10000
8000
6000
4000
2000
0



Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM3')

```
Reconstruction error: 0.635359
Result: Anomaly detected
Reconstruction error: 0.640902
Result: Anomaly detected
Reconstruction error: 0.640783
Result: Anomaly detected
Reconstruction error: 0.640781
Result: Anomaly detected
Reconstruction error: 0.640621
Result: Anomaly detected
Reconstruction error: 0.640626
Result: Anomaly detected
Reconstruction error: 0.640676
Result: Anomaly detected
Reconstruction error: 0.640589
Result: Anomaly detected
Reconstruction error: 0.640554
Result: Anomaly detected
Reconstruction error: 0.640656
Result: Anomaly detected
Reconstruction error: 0.635160
Result: Anomaly detected
```

```
Reconstruction error: 0.048585
Result: Normal activity
Reconstruction error: 0.048608
Result: Normal activity
Reconstruction error: 0.048602
Result: Normal activity
Reconstruction error: 0.048615
Result: Normal activity
Reconstruction error: 0.048624
Result: Normal activity
Reconstruction error: 0.048636
Result: Normal activity
Reconstruction error: 0.048620
Result: Normal activity
Reconstruction error: 0.048614
Result: Normal activity
Reconstruction error: 0.060282
Result: Normal activity
Reconstruction error: 0.048
```

https://studio.edgeimpulse.com/public/999040/live

Part 1 Q:

1. Raw Inputs Three-axis accelerometer data: accX, accY, and accZ, sampled at 100 Hz with a 2,000 ms window size and a 220 ms window increase (stride).

2. NN Classifier Features

- Type: Spectral features, generated by the Spectral Analysis processing block
- Number of input features: 33 (11 per axis across accX, accY, accZ)
- Five feature names used to train the classifier:
    1. accX RMS
    2. accX Peak 1 Height
    3. accZ Spectral Power 2.0–5.0 Hz
    4. accY Spectral Power 2.0–5.0 Hz
    5. accX Peak 1 Freq

3. NN Classifier Outputs 2 output classes: nominal and off

4. Anomaly Detector Features

- Type: Spectral features (same block as the NN classifier)
- Number of features: 33
- Feature names: accX RMS, accX Peak 1 Freq, accX Peak 1 Height, accX Peak 2 Freq, accX Peak 2 Height, accX Peak 3 Freq, accX Peak 3 Height, accX Spectral Power 0.1–0.5 Hz, accX Spectral Power 0.5–1.0 Hz, accX Spectral Power 1.0–2.0 Hz, accX Spectral Power 2.0–5.0 Hz — and the identical 11 features repeated for accY and accZ

- Output: 1 value — Anomaly score

5.

```
Total Size   : 1024KB
Planes       : 1
Lock Regions : 0
Locked       : none
Security     : false
Erase flash

Done in 0.001 seconds
Write 306264 bytes to flash (75 pages)
[==============================] 100% (75/75 pages)
Done in 12.544 seconds
New upload port: COM3 (serial)
Flashed your Arduino Nano 33 BLE development board
To set up your development with Edge Impulse, run 'edge-impulse-daemon'
To run your impulse on your development board, run 'edge-impulse-run-impulse'
Press any key to continue . . .
```

```
Sampling...
Timing: DSP 44 ms, inference 610 us, anomaly 526 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.010460
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 610 us, anomaly 510 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.012533
Starting inferencing in 2 seconds...
```

6.

The classifier is predicting "off" with 56% confidence and the anomaly score is extremely high at 16.0. The results should not always be trusted for several reasons:

- The confidence between nominal and off is close (0.44 vs 0.56), meaning the classifier is not certain
- The high anomaly score of 16 suggests the current sensor data looks very different from anything seen during training, meaning both outputs are unreliable when the anomaly score is this high
- The model was trained on a specific person's ankle-worn sensor data, but is now running on a bench-mounted Arduino with a different IMU (BMI270 vs the original dataset sensor)

When to trust it: Only when the anomaly score is low (close to 0) and one class has clearly higher confidence (above ~0.8). A high anomaly score is a signal that the input is out-of-distribution and the classifier output should be treated with skepticism.

Threshold chosen: 0.508632 , the 95th percentile of reconstruction errors on normal training windows. This ensures 95% of normal windows are accepted as normal while flagging higher-error windows as anomalies.

Reconstruction error distinguishing normal vs anomaly: The autoencoder is trained only on normal data, so it learns to reconstruct normal patterns well (low error). Anomalous activities have unfamiliar patterns that the autoencoder reconstructs poorly, producing higher error above the threshold.

Information that must be preserved from notebook to Arduino: The threshold value, the int8 quantization parameters (input/output scale and zero point), the window size (100 samples), and the flattened input format (100×3 = 300 features).

Why preprocessing must match: If the Arduino uses a different window size, sampling rate, or input format than the notebook, the model receives data it was never trained on, making all predictions meaningless regardless of model quality.

Main limitations on microcontroller:

- Int8 quantization introduces rounding error that shifts reconstruction errors higher, requiring threshold recalibration after deployment
- No retraining on-device, so the model cannot adapt to new normal behaviors
- The model is specific to the training subject and sensor, different users or IMUs produce out-of-distribution data
- Fixed window size and stride means it cannot handle variable-speed or irregular activities well
- High anomaly scores make classifier outputs untrustworthy even when a class prediction is returned